

MATH 394 GEOM ADV VIEWPOINT I
CRN 16740 FALL 2026

1 GENERAL INFORMATION

1.1 Classroom and Meeting Time 119 Fenton Hall; MWF:12:00–12:50pm

1.2 Text Book John Lee, Axiomatic geometry. AMS, 2013

1.3 Instructor Peng Lu

1.4 Office Hours MWF, 2:00pm–2:50, or by appointments

1.5 Office and Phone Number Univ Hall 302, 541-346-4727

1.6 Email Address penglu@uoregon.edu

1.7 Web Page <https://canvas.uoregon.edu/>

1.8 Important dates

- Midterm: Wednesday, October 28, in class
- Veterans Day holiday: Monday, November 11th, no class
- Thanksgiving holidays: Thursday, November 26th and Friday, November 27th
- Final Exam: Final exam schedule will be determined by Registrar. Please check twice during the term, in particular, in 9 and 10 of the term

1.9 Homework There will be about seven write-up assignments. Each will be due on class on the Wednesday of the following week after it is assigned.

Group work on homework is encouraged, but each student must individually write and turn in her/his own assignment. Please do not ask AI to do these homework for you.

It is important that you carefully write out homework solutions on paper in a legible manner, presenting your work in a clear, logical, and orderly way, and using correct notations. Note that having the correct final answer in exams is not sufficient for full credits, the instructor is more interested in assessing your ability to correctly write complete explanations of the solutions demonstrating your understanding of the material.

1.10 AI homework I intend to give two AI-related homework for the quarter. I will discuss with you in the first two weeks about the details.

Separately for this course you are encouraged to download and develop familiarity with some software for exploring geometry on a computer, in particular, drawing figures to enhance your intuition.

1.11 Oral presentations in groups Students will be giving class presentations every Friday. The context is about proofs from the homework, or reading some articles, or reading a few pages of the text book, etc. The format is to divide the class into groups of 4 or 5 students (fixed for the quarter), and each group has 10 to 15 minutes to present by a group representative.

You will earn your grade by being prepared and by participating in the class discussions.

1.12 Attendance Attendance is not required but expected, but a significant portion of your grade (15%) will come from attending and participating on Friday's class of oral presentations by students. You should plan to attend every Friday's class, in the case when you are unable to be present, please let me know in advance (if possible).

1.13 Exams One midterm and one final exam.

You must bring UO photo ID to the midterm and the final.

Let me know asap if you have to miss an exam.

Makeup exams are allowed only for an extraordinary circumstance. In the event of an such a circumstance, I will work with you to consider how we might make adjustments to course requirements where possible. Please let me know as soon as you are able if you experience an extraordinary event. There is no need to explain or offer personal information about the nature of your emergency — I will trust you only to activate this policy in an emergence situation.

1.14 Grade HW 25%, AI HW 10%, Oral presentations 15%, Midterm 25%, and Final 25%
Final letter grade: A: 90% or more, B: 80%–89%, C: 70%–79%, D: 60%–69%, F: below 60%.

Plus grades will be awarded when the last digit is 8 or higher. Minus grades will be given if the last digit is either a 0 or 1. Your final percentage will be rounded to the nearest whole number. You must get at least 70% to receive a pass grade (P) if you are taking this course with the Pass/No Pass option.

2 COURSE GOAL

2.1 Course plan This course is to give a rigorous treatment of the idea of axiomatic geometry, culminating the Hilbert's axioms for Euclidean geometry in two dimensions and some introduction to hyperbolic geometry. In the process the role of parallel postulate and the analysis of problems in geometry using synthetic approach will be emphasised. Basically we will cover Chapter 2, 3 6, and part of Chapter 5, 7, 10, 18 of the book by J. Lee.

Throughout the course we will emphasizes investigations, proofs, and challenging problems.

2.2 Learning outcomes Students successfully completing this course will be able to:

- Give clear and concise written proofs of standard geometry propositions.
- Give verbal presentations of Euclidean geometry proofs in front of a group of people.
- Give examples of axiomatic systems, and models of axiomatic systems.
- Know the statements, and be familiar with the proofs, of several advanced theorems of Euclidean geometry. Among these are theorems about circles circumscribed and inscribed in triangles, the circumcenter, and the centroid associated to a triangle, and the theory of cyclic quadrilaterals. Moreover, students should be able to use these results in the proofs of other theorems.

Weekly homework, AI homework, reading and oral presentations, as well as problems on the midterm and the final exam, will provide students with opportunities to demonstrate the level of their abilities relative to the above learning outcomes.

3 OUTLINE AND HOME WORK SCHEDULE

The following are the tentative schedule, we may adjust it as the term moves.

Week 1: September 28 to October 02 (Chapter 2) Homework 1 assigned

Week 2: October 05 to 09 (Chapter 2) Homework 2 assigned

Week 3: October 12 to 16 (Chapter 3) Homework 3 assigned

Week 4: October 19 to 23 (Chapter 3) Homework 4 and 5 (combined) assigned

Week 5: October 26 to 30 (Chapter 4) Review for Midterm on Monday and
Midterm on Wednesday

Week 6: November 02 to 06 (Chapter 5 and 6) Homework 6 assigned

Week 7: November 09 to 13 (Chapter 6) Homework 7 assigned

Week 8: November 16 to 20 (Chapter 7 and 10) Homework 8 and 9 (combined) assigned

Week 9: November 23 to 27 (Chapter 10 and 18) (Thanksgiving Holidays: 28 and 29)

Week 10: November 30 to December 04 (Chapter 18 and Review for final) No homework

4 SPECIAL ACCOMMODATIONS

Please check out at

<https://teaching.uoregon.edu/university-course-policies>

In particular, about Special Accommodation, The University of Oregon is working to create inclusive learning environments. If you are a student with a documented disability, please meet with me soon to discuss your needs. If you have not already requested a notification letter from Accessible Education Center (AEC) outlining recommended accommodations, please do so soon.